

Single-Wire 4x4 MATRIX KEYPAD Using AVR

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This tutorial demonstrates a different way of interfacing a 4x4 matrix keypad with ATmega32 AVR microcontroller. Keypad is one of the most important input devices used in electronics projects. It is one of the easiest ways to give commands or instructions to an electronic system. But when designing a traditional matrix keypad, you require more number of input/output (I/O) pins. For example, if you are developing a 4x4 keypad of 16 keys, you require eight digital I/O lines of the microcontroller device.

This project uses only a single analogue pin to achieve the same goal, so the remaining pins are available for other purposes.

Fig. 1 shows the block diagram of the single-wire keypad with liquid-crystal display (LCD). The system consists of a 4x4 keypad that inputs data to the microcontroller through a single wire, ATmega32 microcontroller for processing the received signal, power supply section and a 16x2 LCD for displaying the keys.

Circuit and working

Complete circuit diagram of the single-wire 4x4 matrix keypad using AVR is shown in Fig. 2. It is built

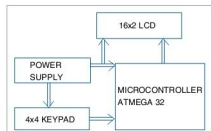


Fig. 1: Block diagram of single-wire 4x4 keypad

around step-down transformer X1, 16 pushbutton switches (S1 through S16), 16 resistors (R1 through R16), a 5V regulator 7805 (IC1), an ATmega32 microcontroller (IC2), an LCD (LCD1) and a few other components. It has three sections: power supply, display and keypad.

Power supply.

The 230V AC mains is stepped down by a 230V AC primary to 9V, 250mA secondary transformer (X1), rectified by a bridge rectifier module (BR1) and filtered by capacitors C1 and C2. The voltage is regulated by 7805 regulator IC. LED1 glows to indicate the presence of power in the circuit.

The regulated 5V DC powers the microcontroller and LCD1. You can also use a 9V AC mains adaptor to drive the circuit.

Switch S17 is used to power the circuit. Connect switch S17 to 9V if you do not use transformer X1.

Display. The project uses a 16x2, Hitachi HD44780-con-

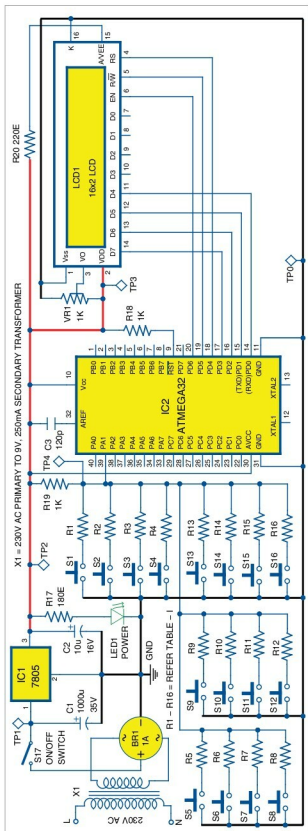


Fig. 2: Circuit diagram of single-wire 4x4 keypad